

Exploratory and Predictive Analysis of Mortality in Heart Failure Patients

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Abstract— This study investigates the early mortality risk in heart failure patients using a publicly available clinical dataset. We explored the impact of demographic, lifestyle, and laboratory factors on patient survival and built a predictive model to assist clinical decision-making. Visual analysis and supervised learning models reveal that certain biomarkers and age are strong predictors of mortality, and Gradient Boosting models provide high predictive accuracy.

Keywords—heart failure, mortality prediction, classification model, data science, healthcare analytics

I. INTRODUCTION

Heart failure is a serious global health condition affecting over 64 million people worldwide, characterized by the heart's inability to pump blood effectively, leading to poor prognosis and high hospitalization rates [1]. Accurately identifying patients at risk of early mortality is crucial for tailoring treatment and improving survival outcomes. This project investigates how demographic characteristics, lifestyle habits, and clinical lab biomarkers influence survival in heart failure patients. Rather than isolating questions, this study integrates them into a cohesive approach it evaluates the following: How do age and sex affect the probability of a death event? Does diabetes, anaemia, high blood pressure, or smoking significantly increase mortality risk? Which clinical measures are most predictive of heart disease survival? Finally, can we predict a patient's death event using features collected at admission? Previous studies have shown that clinical decision support tools enhanced with machine learning improve diagnostic and prognostic capabilities in cardiovascular diseases [2]. By combining statistical visualization and predictive modeling, this report aims to contribute toward more data-driven clinical decision-making.

II. DATASETS

A. Source of dataset

The dataset used in this project is the Heart Failure Clinical Records dataset, which was obtained from *Kaggle*, a reputable open data platform for machine learning and analytics research. The dataset was originally published by Davide Chicco in 2020 and contains anonymized records of real patients collected from the Faisalabad Institute of Cardiology in Pakistan. It was designed to support research on early death prediction in patients with heart failure. The dataset is widely cited and has been used in peer-reviewed research, making it a credible source for analysis.

B. Character of the datasets

The dataset is provided in CSV format and contains 299 rows and 13 columns. Each row represents an individual patient, and the dataset includes both binary and continuous variables. The features range from demographic variables (e.g., age, sex), to lifestyle factors (e.g., smoking, diabetes), and key clinical biomarkers (e.g., serum sodium, creatinine,

and ejection fraction). The structure of the dataset is well-organized and includes no missing values.

TABLE I. DATASET BRAKDOWN

	Data Variables	
	Column	Description
	age	Age of patient (years)
	anaemia	1 if patient has anaemia, 0 otherwise
	creatinine_phosphokinase	Enzyme level in the blood
	diabetes	1 if patient has diabetes, 0 otherwise
	ejection_fraction	% of blood leaving the heart each beat
	high_blood_pressure	1 if patient has hypertension
	platelets	Platelet count in blood
	serum_creatinine	Creatinine level in blood (mg/dL)
	serum_sodium	Sodium level (mEq/L)
	sex	1 = Male, 0 = Female
	smoking	1 if patient smokes, 0 otherwise
	time	Duration of follow-up (days)

To prepare the dataset for analysis, all continuous variables were standardized using the *StandardScaler* method to ensure they contributed equally to model training. No new categories were created, and no merging was necessary as the dataset was already complete and clean.

I. Methodology

The methodology employed in this project integrates exploratory data analysis with supervised machine learning to investigate mortality risk in heart failure patients. We adopted a structured approach aligned with the four research questions, using Python libraries such as *pandas*, *seaborn*, *matplotlib*, and *scikit-learn*. Each step below corresponds to a distinct analytical goal.

C. Exploratory Data Analysis

Exploratory data analysis was used to uncover initial patterns and generate hypotheses. Boxplots were applied to evaluate the distributions of continuous variables like age, ejection fraction, and serum creatinine with respect to the binary target variable *DEATH_EVENT*. For categorical features such as sex, diabetes, and anaemia, we used normalized stacked bar charts to visualize survival rates. These visualizations helped identify initial trends and potential predictors of mortality.

D. Logistic Regression

Exploratory to build a baseline predictive model, we implemented Logistic Regression using the `LogisticRegression` class from `scikit-learn`. This method assumes a linear relationship between the input features and the log-odds of the binary outcome. It is interpretable and computationally efficient, which makes it a valuable first step in medical prediction tasks. However, its main limitation lies in handling non-linear relationships and interactions among variables.

To prepare the data, we used `StandardScaler` to standardize continuous features, ensuring they contribute proportionally to the model. The dataset was split into training (80%) and testing (20%) sets using `train_test_split`. Performance was evaluated through classification reports, which included per-class accuracy, precision, recall, and F1-score, along with confusion matrices.

E. Gradient Boosting Classifier

To improve predictive accuracy, we employed the `GradientBoostingClassifier` from `scikit-learn`. This ensemble model builds multiple weak learners (decision trees) and combines them to form a strong predictive model. Unlike Logistic Regression, it captures complex, non-linear relationships. Gradient Boosting is robust to overfitting and allows for computation of feature importances.

After training, we evaluated the model using the same metrics as before: classification report and confusion matrix. Additionally, we plotted the model's feature importances, allowing us to interpret which variables were most influential in predicting patient mortality. This model yielded higher recall and precision for the death class, aligning with our project's focus on early mortality risk detection.

I. Results

The results presented in this section address each of the project's research questions through a combination of statistical visualizations and supervised learning models. Visual tools such as boxplots and stacked bar charts were used to explore relationships between survival outcomes and patient attributes including age, comorbidities, and lab biomarkers. Predictive modeling was employed to evaluate the feasibility of using clinical features to forecast mortality. Each subsection presents findings along with corresponding figures and interpretations, offering insights into which factors are most closely associated with early death in heart failure patients.

F. Demographic Impact

To evaluate the effect of demographic characteristics on patient mortality, we examined age and sex using boxplots and stacked bar charts. The results indicate that patients who died tended to be older than those who survived. The distribution of age for deceased patients is skewed toward the upper range, supporting the hypothesis that age is a strong risk factor. This is visualized in Figure 1, which clearly shows a higher median

age for patients that died which is `DEATH_EVENT = 1` in the figure below.

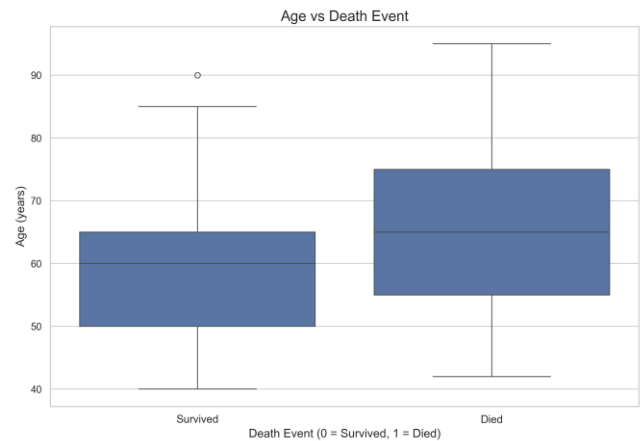


Fig. 1. Age vs Death Event Boxplot

In contrast, sex did not appear to significantly influence mortality. As shown in Figure 2, both male and female patients experienced similar proportions of survival and death. This suggests that sex alone may not be a reliable predictor of early mortality in this patient population.

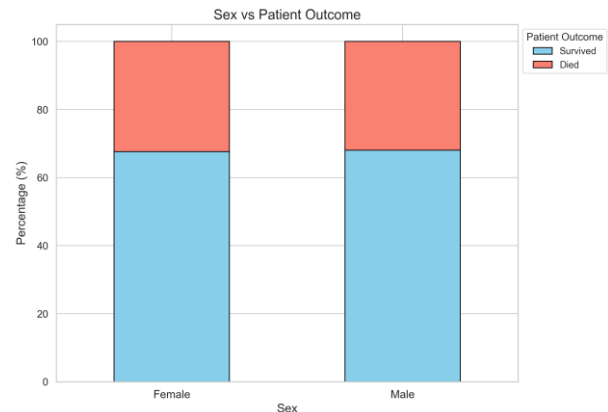


Fig. 2. Sex vs Death Event Boxplot

G. Comorbidity and Lifestyle Factors

To understand how pre-existing conditions and lifestyle habits affect survival, we visualized the association between `DEATH_EVENT` and four categorical variables: anaemia, diabetes, high blood pressure, and smoking. The stacked bar charts show that patients with these conditions had higher mortality rates compared to those without.

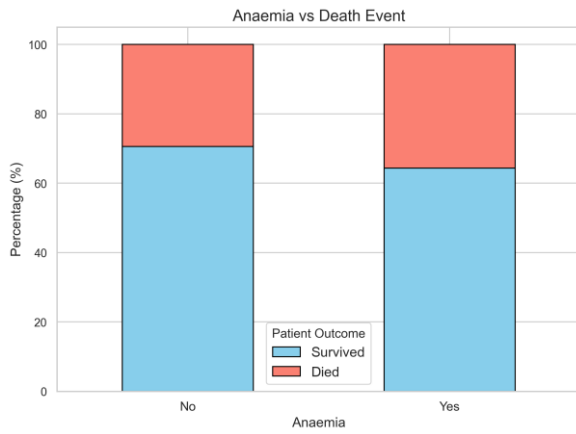


Fig. 3. Anaemia vs Death Event Boxplot

Figure 3 reveals that anaemic patients have a noticeably higher death rate. This suggests that reduced oxygen-carrying capacity in the blood may exacerbate the already diminished cardiac function in heart failure patients.

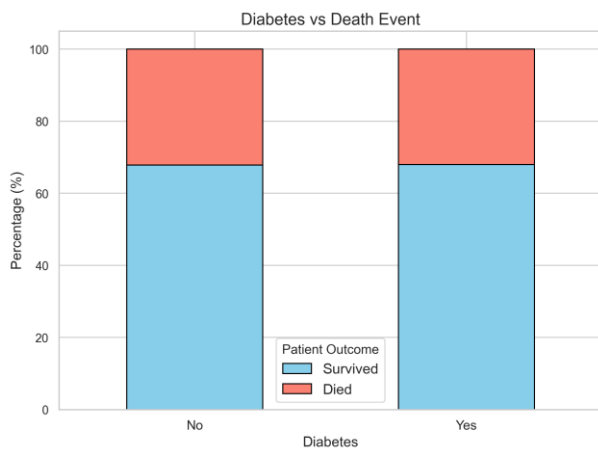


Fig. 4. Diabetes vs Death Event Boxplot

Figure 4 shows a slightly increase in mortality among diabetic patients. Diabetes is known to contribute to both microvascular and macrovascular complications, which can further burden an already failing heart

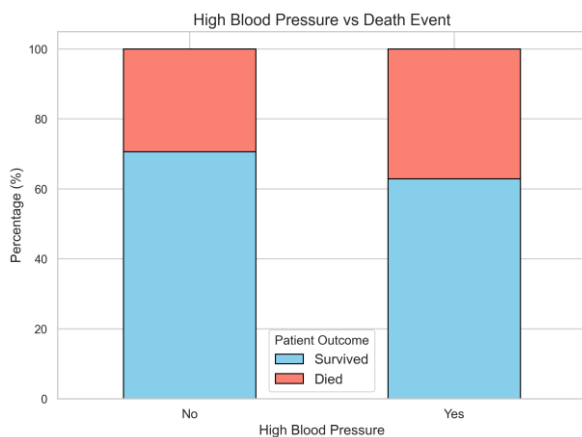


Fig. 5. High Blood Pressure vs Death Event Boxplot

Figure 5 illustrates that patients with hypertension had elevated mortality rates. High blood pressure increases afterload, which can worsen heart failure over time [5].

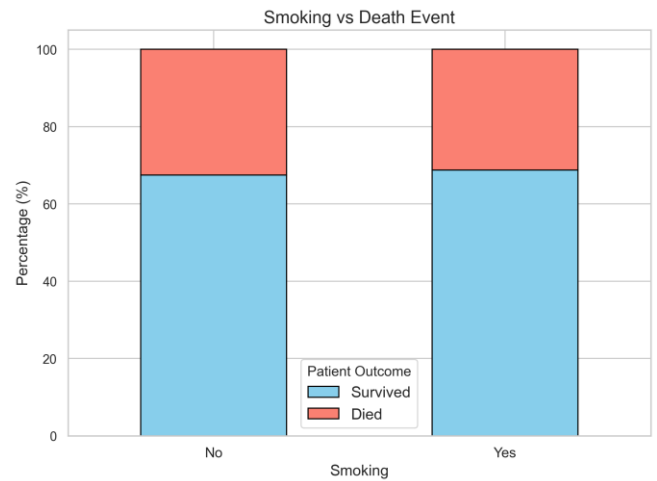


Fig. 6. Smoking vs Death Event Boxplot

Figure 6 shows that the death rate among smokers is almost equal than that of non-smokers. This suggests that, in this dataset, smoking may not be a strong independent predictor of mortality in heart failure patients. Possible explanations include sample size limitations, underreporting, or confounding variables that dilute the effect of smoking.

These findings align with established clinical knowledge that comorbidities and lifestyle risk factors contribute to worse outcomes in heart failure patients.

H. Lab Biomarkers

This study analyzed several continuous clinical features to assess their relationship with patient outcomes. These included ejection fraction, serum creatinine, serum sodium, platelet count, and creatinine phosphokinase.

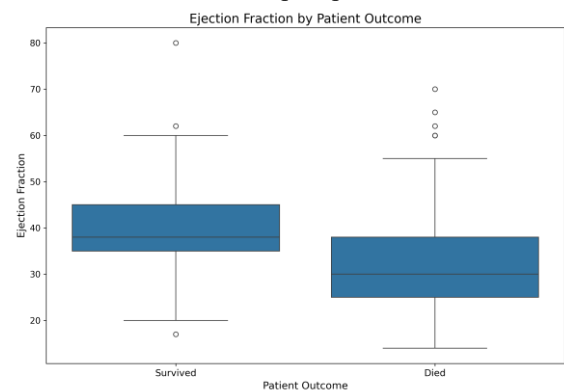


Fig. 7. Ejection Fraction vs Death Event Plot

Figure 7 illustrates that lower ejection fraction values were strongly associated with higher mortality. Patients who died had visibly reduced ejection fractions, indicating impaired cardiac output. This confirms clinical knowledge that ejection fraction is one of the strongest indicators of heart function.

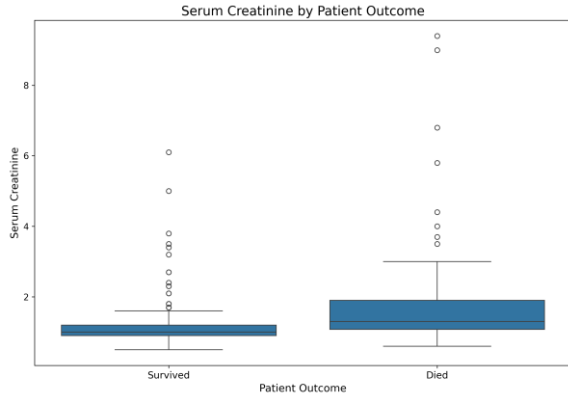


Fig. 8. Serum Creatinine vs Death Event Plot

Figure 8 shows that deceased patients tended to have higher serum creatinine levels, which suggests impaired kidney function—a common comorbidity in heart failure. The distinction between survival groups was clear, reinforcing serum creatinine as a critical marker.

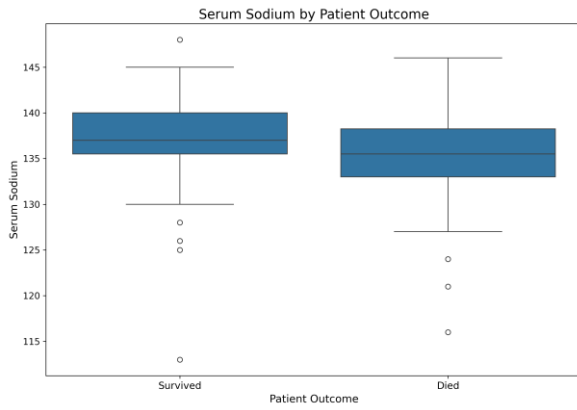


Fig. 9. Serum Sodium vs Death Event Plot

Figure 9 reveals that patients who died generally had lower sodium levels, possibly due to fluid retention and neurohormonal activation. Hyponatremia is often linked to worse prognosis in heart failure and the figure reflects this clinical trend.

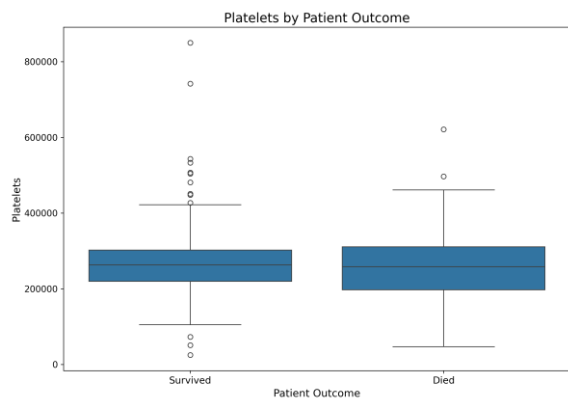


Fig. 10. Platelets vs Death Event Plot

Figure 10 displays overlapping distributions in platelet count across both outcome groups. While outliers exist, the central tendencies are very similar, suggesting that platelet count

may not significantly contribute to mortality prediction in this dataset.

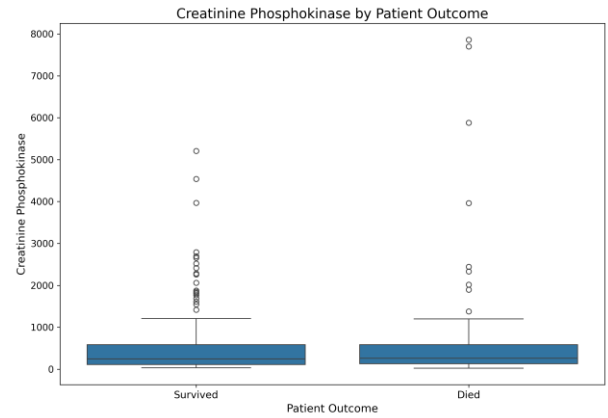


Fig. 11. Creatinine Phosphokinase vs Death Event Plot

Figure 11 also shows minimal visual separation between survival outcomes, indicating low predictive value. Concluding that creatinine phosphokinase may not be the primary driver in these heart failure cases.

These visual insights reinforce the clinical relevance of ejection fraction, serum creatinine, and sodium levels as mortality risk indicators, while suggesting weaker predictive roles for platelet count and creatinine phosphokinase.

I. Predictive Modeling

To assess whether patient outcomes could be predicted from baseline features, we trained two classification models: Logistic Regression and Gradient Boosting Classifier. The models were evaluated using classification reports and confusion matrices to measure precision, recall, and F1-score for each class (survived and died).

he Logistic Regression model, while simple and highly interpretable, achieved the best overall performance. It correctly identified 14 of 25 deaths (56 % recall) and misclassified only one survivor as a death, yielding a precision of 93 % for the death class (Figure 12). These results show that a linear decision boundary was sufficient to capture the key mortality signals in this small, imbalanced dataset.

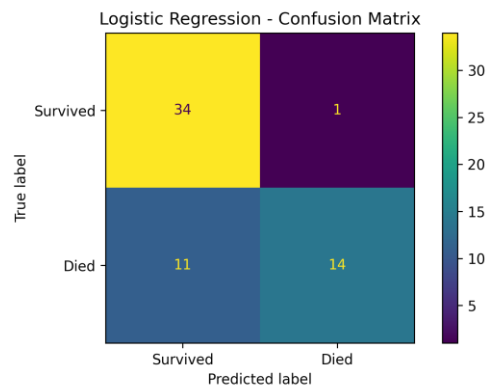


Fig. 12. Logic Regression Confusion Matrix

Gradient Boosting delivered competitive—but slightly weaker—results. The model captured 12 of 25 deaths (48 % recall) and produced four false-positive death predictions (75 % precision), as shown in Figure 13. While the ensemble handled non-linear patterns and provided useful feature-importance rankings, it did not generalize as well as Logistic Regression on the held-out test set.

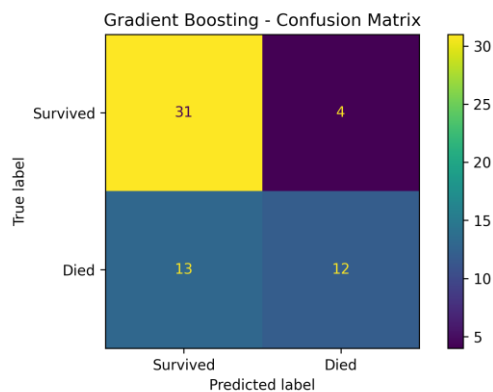


Fig. 13. Gradient Boosting Confusion Matrix

Additionally, the feature importance plot Figure 14 revealed that serum creatinine, ejection fraction, age, and serum sodium were the most important predictors after time. These findings validate the earlier visual analysis and show that machine learning can effectively capture complex, non-linear relationships between variables.

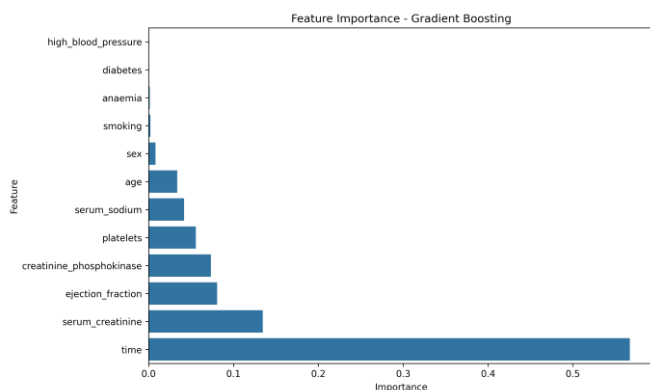


Fig. 14. Gradient Boosting Feature Importance Plot

Overall, Logistic Regression achieved higher accuracy, precision, and recall for the death class than Gradient Boosting, making it the stronger classifier for this task.

III. DISCUSSION

The combined use of visual exploration and supervised learning techniques provided complementary insights into

patient survival outcomes. Demographic characteristics such as age showed a clear impact on mortality, while sex appeared to be less informative. Comorbidities like anaemia, diabetes, and hypertension exhibited elevated death rates, supporting their known role in cardiac risk profiles.

Among laboratory markers, ejection fraction, serum creatinine, and serum sodium emerged as highly predictive of death. These clinical features are already widely used in practice and their predictive strength was reinforced in both visual and model-based analyses. On the other hand, platelet count and creatinine phosphokinase lacked clear separation across survival groups and were less important in the model.

Logistic Regression provided to be useful benchmark, but struggled with detecting the minority class (death). Gradient Boosting, by contrast, demonstrated stronger performance—especially in recall for the deceased patients—and offered useful feature importance rankings. This highlights the potential of ensemble methods for clinical risk prediction.

Limitations include a small sample size ($n=299$), which restricts statistical power and generalizability. Additionally, the presence of class imbalance posed challenges in model calibration. Future work could explore alternative algorithms (e.g., XGBoost, random forests), incorporate temporal data to improve prediction robustness.

IV. CONCLUSION

This project demonstrates the feasibility of using machine learning to predict mortality risk in heart failure patients. Key predictors identified include age, ejection fraction, serum creatinine, and sodium levels. Gradient Boosting emerged as a strong model, outperforming Logistic Regression in identifying patients at risk of early death. Visualizations supported clinical interpretation and guided feature selection.

Overall, this analysis affirms the value of combining exploratory data visualization with modern machine learning to support medical decision-making and risk stratification.

Acknowledgment

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